

The Power of Vaccination:

From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of two centuries of vaccine innovation, disease elimination, and the remaining challenges for global public health • 1796 – 2020

For: Public Health Professionals & Policy Analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Key insight

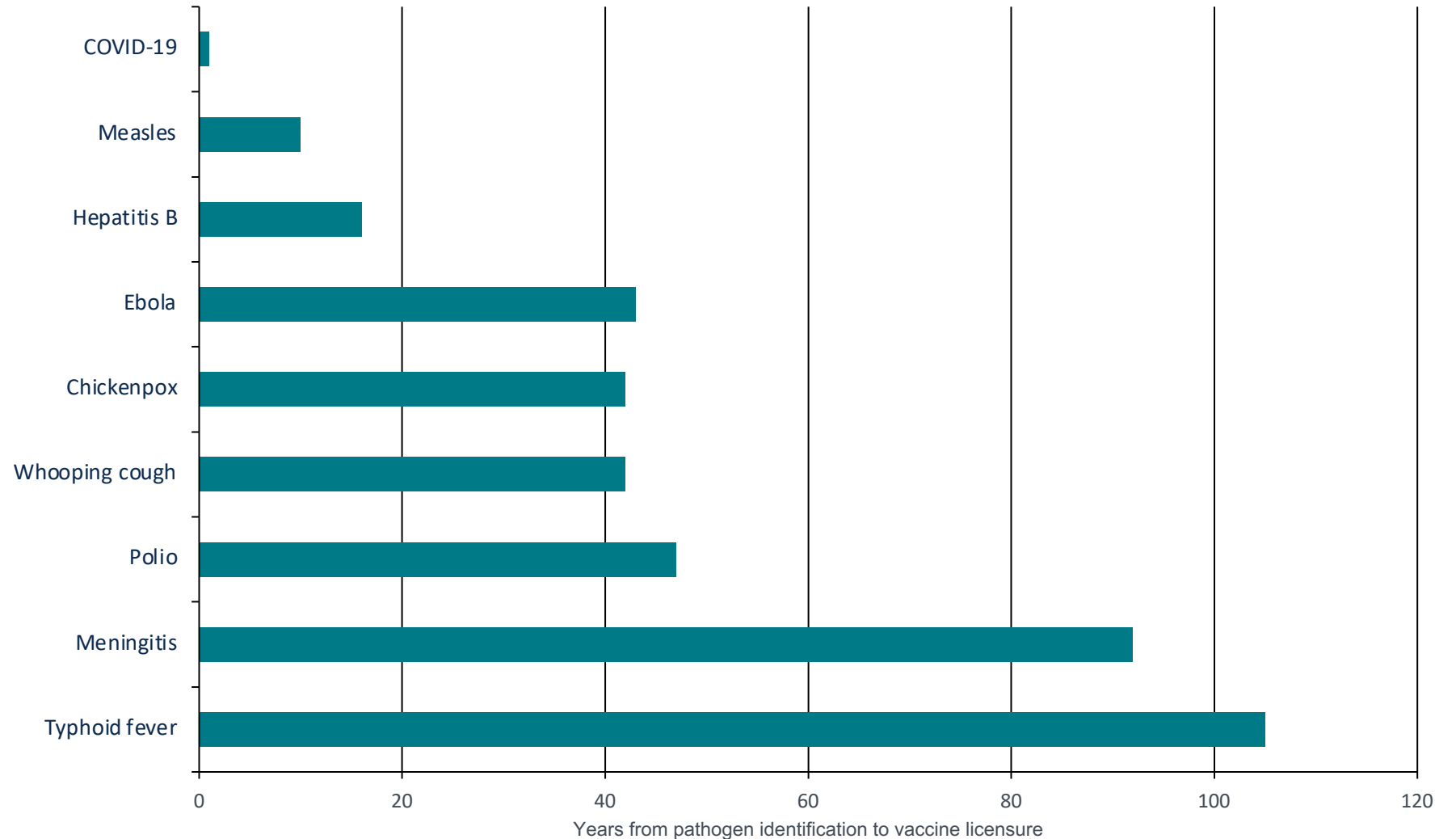
Typhoid fever: 105 years

Meningitis: 92 years

Polio: 47 years

COVID-19: < 1 year

mRNA technology & genomic tools collapsed the timeline.



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

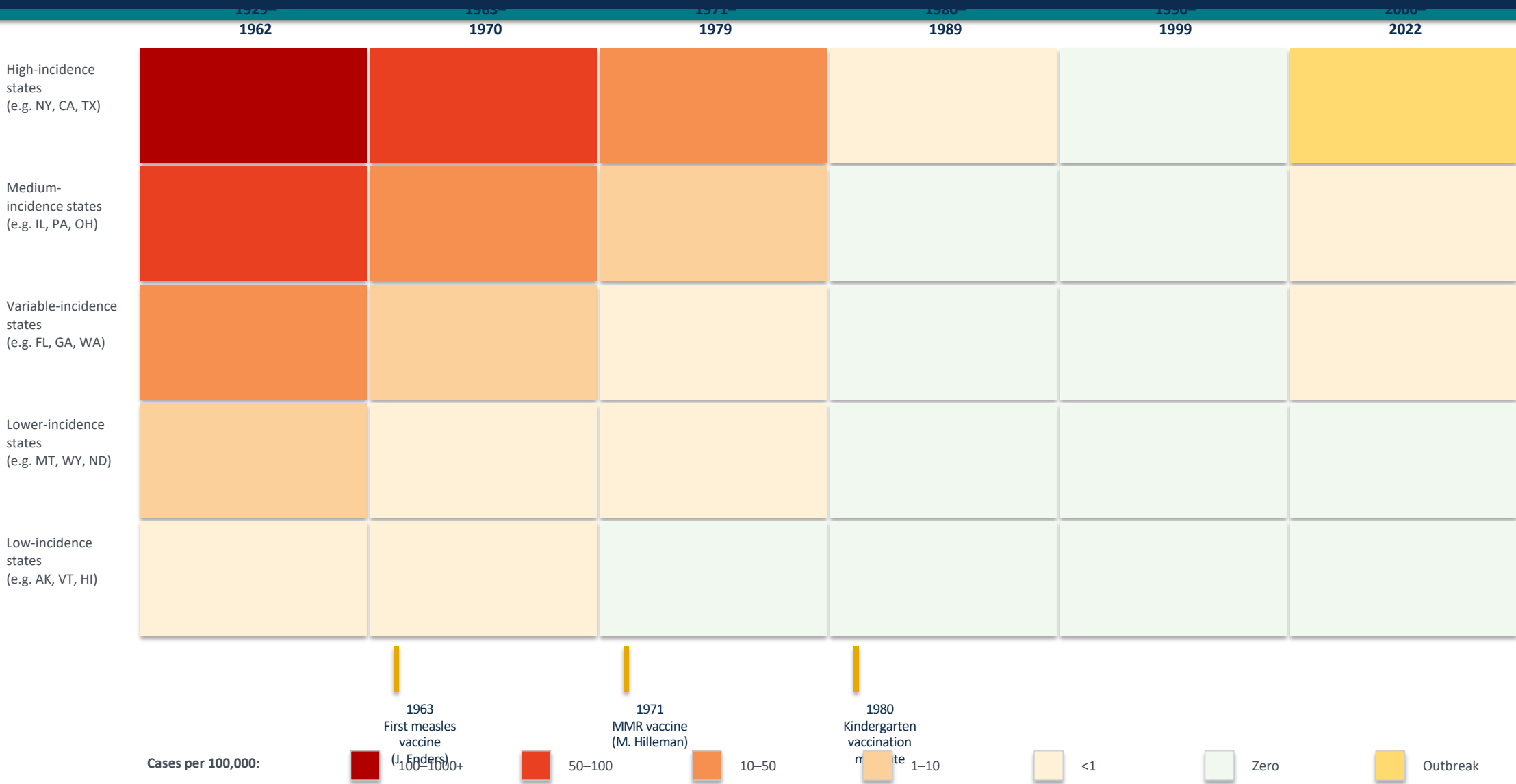
Disease	Pre-Vaccine Cases/M/yr	Post-Vaccine Cases/M/yr	Case Reduction	Pre-Vaccine Deaths/M/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute Hep B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%



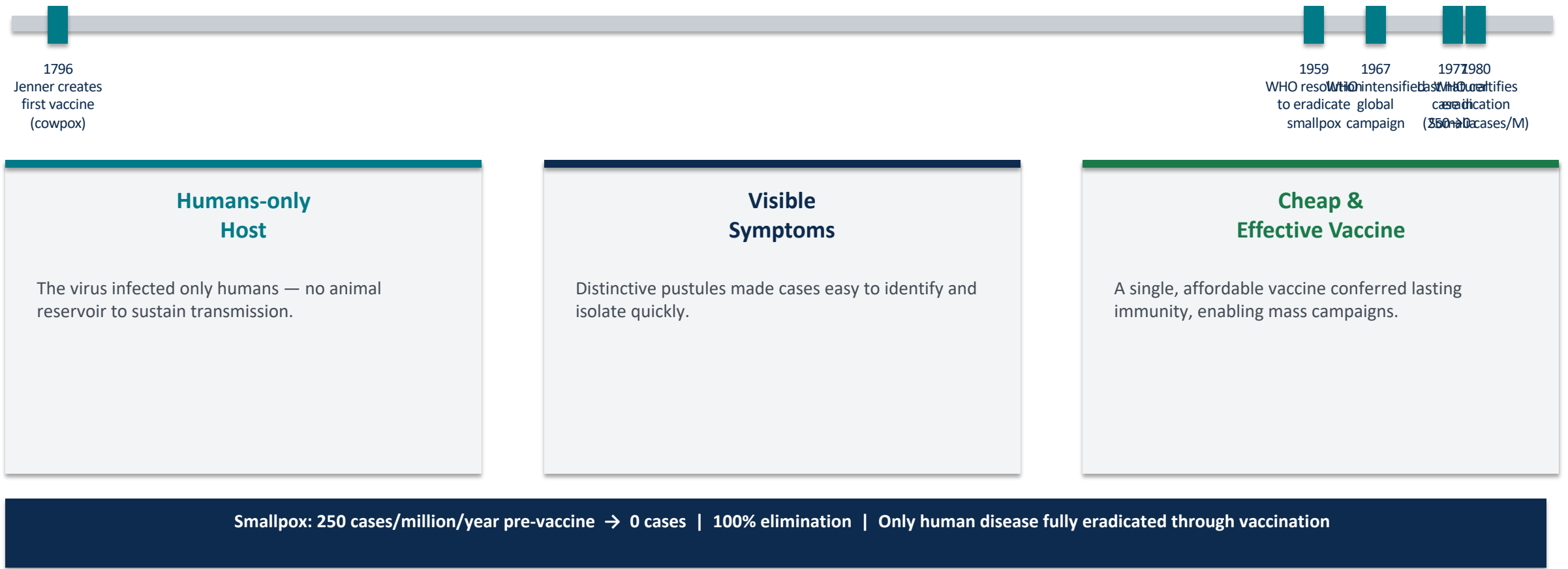
= 100% elimination of both cases and deaths

Cases & deaths per million population per year | Data: Roush & Murphy (2007)

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



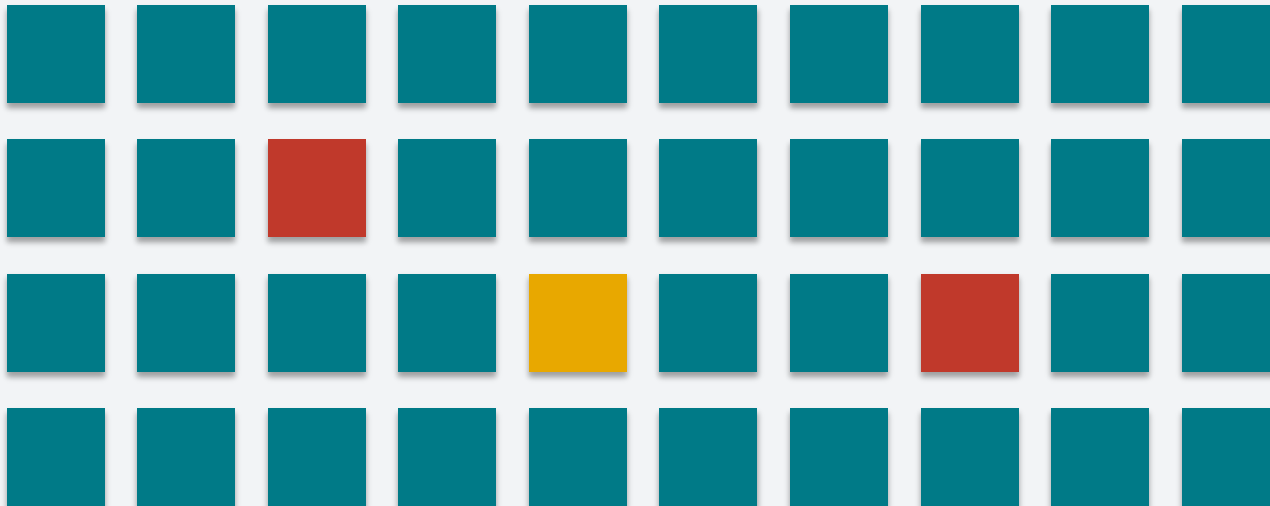
Smallpox: The Only Human Disease Eradicated Through Vaccination



Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Community Vaccination Shield

-  Vaccinated individual (protected)
-  Unvaccinated — susceptible
-  Cannot be vaccinated (vulnerable)
-  Pathogen / exposure source



 Pathogen enters →
blocked by vaccinated
population shield

How it works

Shield effect

When enough people are vaccinated, the pathogen cannot spread effectively through the community — even reaching those who are unvaccinated.

Who is protected?

Newborns too young for vaccination, immunocompromised individuals, and cancer patients on chemotherapy who cannot safely receive vaccines.

Why coverage matters

Gaps in vaccination coverage create pathways for pathogens to spread, putting the most vulnerable at disproportionate risk.

Vaccines protect not only individuals — they shield entire communities.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal Disease

40.5%

Case reduction

31.3%

Death reduction

Significant burden remains; lower vaccine uptake and newer serotypes.

Hepatitis A

89%

Case reduction

88.7%

Death reduction

Good progress, but access gaps persist in lower-income regions.

Acute Hepatitis B

83.9%

Case reduction

83.6%

Death reduction

Mother-to-child transmission and access remain key barriers.

Systemic Barriers to Full Coverage

Supply Shortages

Supply chain vulnerabilities leave children in low-income countries at risk when stocks run out.

Access Gaps

Globally, not everyone has physical or financial access to vaccines, leaving communities unprotected.

Trust & Uptake

Poor scientific understanding and misinformation reduce trust and uptake even where vaccines are available.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

100% elimination achieved

Diphtheria, smallpox, and acute polio have been 100% eliminated in the US in both cases and deaths. Measles is at 99.99% reduction.

Timelines collapsed

Vaccine development went from 105 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic sequencing tools.

Herd immunity protects all

Community vaccination shields those who cannot be vaccinated — newborns, immunocompromised patients, and chemotherapy patients.

Gaps persist

Pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%) still require improved vaccines, access, and coverage.

Call to action

Health systems must invest in supply chain resilience, equitable access, public trust, and next-generation vaccines to close remaining gaps.